

ROUND 1 / EVIDENCE PACKET

Forecast dust. Control the edge.

A working software prototype for an inspectable construction-dust forecast and proposed zone-based misting control.

What is implemented	What it establishes
Frozen trained model + local inference	A preliminary laboratory monitor forecast
Measured-data replay + honest baselines	Past input, model output and later recorded actual
Four controllers in five common scenarios	Simulated water/exposure tradeoffs under stated assumptions
Original website + offline saved backup	An integrated, reviewable Round 1 demonstration

Why this problem

World Bank reporting identifies major construction with traffic among studied pollution settings in Dhaka. This motivates the site-monitoring proposal; it does not prove our system reduces pollution or improves health. [World Bank source \[1\]](#).

Proposed contribution

Compare forecast-guided zone selection with continuous and reactive control, while exposing the forecast input, failures, water accounting and controller assumptions.

Round 1 boundary

No physical hardware has been built or purchased. Outdoor boundary accuracy and physical misting effectiveness remain unvalidated. Hardware starts only with the team's explicit Round 2 instruction.

Team CTRL_V

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Team ID: not issued by organizers according to the team. Contact details and qualification/feedback are not supplied.

01 / MEASURED FORECAST

A real model. A mixed test result.

Askarov and Choi (2024), Mendeley V1, DOI 10.17632/7f22n9v7hp.1, CC BY 4.0. Twelve laboratory OPC-N3 recordings; raw PM10. [Data source \[2\]](#).

Causal one-second grid; observation age at most 1.5 s. Each forecast uses 121 snapshots over the previous 120 s and predicts the latest-observed snapshot 30 s ahead. Sixteen PM10-only lag/statistic features; no event clock, drilling label or future wind as learned input.

Whole groups 1/2 train (24,818 windows), 3 selects (12,113), 4 tests (15,065). The test group is labelled increased temperature. Missing calendar dates mean this is a group holdout, not strict global chronology.

Predictor	Test MAE (ug/m3)	Test RMSE (ug/m3)
Persistence	95.702	199.384
Trailing mean	81.565	200.713
Trained boosting model	88.405	179.272

Model MAE is 7.62% below persistence and 8.39% above the trailing mean. The mean is the stronger MAE baseline. No test-driven refit or retuning was done. Error improvement is not pollution reduction.

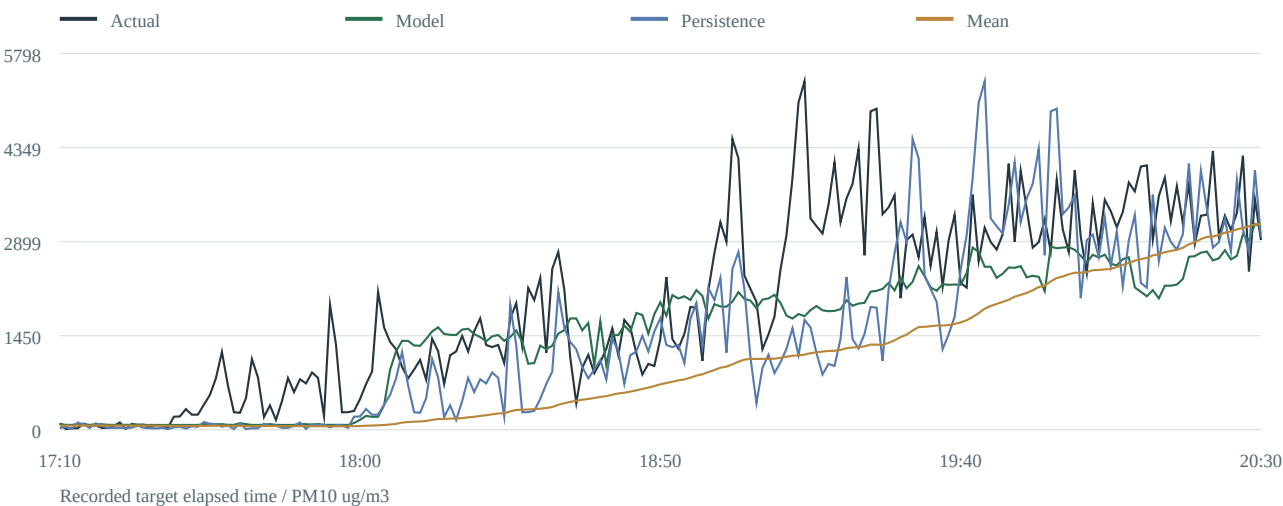


Figure: target-aligned held-out 90-second-label window; each forecast was issued 30 s before its target. Full profiles and failure windows are included in reports/evaluation. Overlapping windows are correlated within only three records. Abrupt rises/peaks and the ten-second-label record are weaknesses.

At a predeclared illustrative 500 ug/m3 warning setting: model 13/18 matched crossing runs, 15 false alerts; mean 13/18, 8 false alerts. Only one of three first onsets receives advance learned warning. A 30-second horizon is not reliable 30-second warning lead.

02 / COMMON SITE EXPERIMENT

One environment. Four control strategies.

All cases use the same source/weather for every controller, equal actuator limits, a 480-second window and four boundaries. PM10 units are ug/m3; the mean is the time average of the maximum boundary each second. Settings are illustrative, not regulatory.

Case / strategy	Water (L)	Mean max PM10	Above 250 (s)	Switches
low-risk / No control	0.000	56.4	0	0
low-risk / Continuous	16.000	45.8	0	4
low-risk / Reactive	0.000	56.4	0	0
low-risk / Predictive	0.000	56.4	0	0
east / No control	0.000	294.6	203	0
east / Continuous	16.000	129.2	86	4
east / Reactive	1.250	176.0	122	4
east / Predictive	1.842	150.0	86	2
diagonal / No control	0.000	172.4	139	0
diagonal / Continuous	16.000	86.4	0	4
diagonal / Reactive	1.083	145.9	68	12
diagonal / Predictive	2.517	108.8	0	4
wind-shift / No control	0.000	281.2	196	0
wind-shift / Continuous	16.000	124.4	86	4
wind-shift / Reactive	1.283	173.9	134	6
wind-shift / Predictive	2.025	143.8	86	4
data-loss / No control	0.000	294.6	203	0
data-loss / Continuous	16.000	129.2	86	4
data-loss / Reactive	2.000	176.0	122	10
data-loss / Predictive	2.467	157.0	104	10

Continuous has the lowest mean concentration and highest water use. Predictive uses less water than continuous but more than reactive in the higher-risk cases. The frozen cases include low risk, eastward plume, two boundaries, wind shift and identical sensor loss.

Assumptions: 120 x 80 m site; background 40; wind 3 m/s; flow 0.5 L/min/zone; incoming construction fraction removed 0.65; actuator delay 5 s; minimum on/off 15 s. Background is preserved. Water integrates commanded flow. Source proxy is concentration, not an emission rate.

The learned source endpoint is mapped through an assumed linear path, held-current wind and untreated boundary rollout. This transfer is uncalibrated and does not establish physical suppression or field boundary accuracy.

Show the input. Reveal the outcome.

Step	Judge demonstration
Measured replay	Observe current PM10, the model's +30 s endpoint and both baselines. Open the 121-input/16-feature detail and artifact identity.
Time catches up	Advance the recorded clock to reveal an earlier forecast's recorded target. Future actual is never supplied to the model.
Site experiment	Diagonal case at 100 s: A north and B east are exposed. Compare strategies at the same clock, then inspect the full eight-minute table.
Recovery	Local live mode executes the fitted Python model without internet after setup. Saved fallback shows previously computed results and disables new assumption runs.

Verification evidence

Twenty focused Python tests pass. Independent simulation checks verify all 20 runs / 9,600 intervals: water, metrics, switching, background, capacity and hashes. Five browser journeys pass: all five fixed fixture predictions to 1e-8, API/display agreement, no early target reveal, playback, stale-response rejection, custom flow, saved/live equality and mobile pages.

Offline rehearsal

A real ten-minute browser journey denies every external network request while using local live inference and saved recovery. Its exact result is in reports/readiness. Physical Wi-Fi is not disabled. A software check does not verify the team's spoken pitch or venue projector.

On the prepared Mac

Double-click Start DustTwin.command. Keep the terminal open and confirm Local model ready. For recovery use Start Saved Replay.command and explicitly identify Saved replay mode.

On another computer

Unzip the release. Saved mode needs only Python 3 and a modern browser; no Node or package install. Live mode requires Python 3.14 and pinned service packages installed once. The archive contains the built website, permitted data/evidence and the frozen model; it does not bundle a Python runtime.

Artifact SHA-256

d78f1b37269f72af45933e01722968fb13ed82178f6d8b3e4c5584d46cec09c7

54,679 bytes / hist_gb_depth3_iter100. Source, configuration, artifact and trace hashes are public. Input snapshot IDs and source attribution are inspectable in the live replay.

04 / PROPOSED ROUND 2

Cost basis.

Calibration before deployment.

Listed prices checked 1 October 2026; not formal quotations, purchased hardware or a complete installed-system cost. References [3]-[9]. USD and BDT remain separate; no exchange rate or payback is invented.

Candidate item	Qty	Listed unit price	Important qualification
Candidate laser PM monitor SEN0177	5	USD 46.90	Listed out of stock. 0–1000 µg/m³ and ≤10 s response differ from training OPC-N3; unsuitable for replay peaks without range/cadence revalidation.
FireBeetle 2 ESP32-E DFR0654	5	USD 8.90	Candidate acquisition/link nodes; the Python model stays on a local laptop/hub, not this microcontroller.
RS485 wind speed SEN0483	1	USD 45.00	Siting, interface adapter and independent calibration required.
RS485 wind direction SEN0482	1	USD 45.00	Align meteorological wind-from with A north/B east/C south/D west.
12 V diaphragm pump set	1	BDT 1360.04	Listing includes adapter/pipe/spray gun, not four calibrated mist heads. Conflicting power/pressure specifications need supplier confirmation.
12 V 3/4-inch normally closed valve	4	BDT 450.05	Listed out of stock. Pressure compatibility and actual operation need confirmation.
Four-channel 12 V relay module	1	BDT 280.04	Logic-level compatibility, isolation, load protection and wiring not verified.

Five-monitor listed subtotal: USD 369.00 plus BDT 3440.28. Three-monitor pilot: USD 257.40 plus BDT 3440.28. Excludes suitable construction-range replacements, mist heads/manifold, flow meter, protection, enclosures, links, reference calibration, labour, shipping/duties/tax.

The proposed low-cost PM sensor differs from OPC-N3: its listed 0-1000 ug/m3 range and up-to-10 s response cannot support all replay peaks or the current one-second task without new data/task validation. It is listed out of stock, as are the valves. Confirm pressure compatibility and relay logic/protection before any physical build.

The pump listing's electrical/pressure specifications conflict. Measure demand, flow and coverage instead of assuming catalogue maxima. Energy = watts x on-seconds / 3,600,000 kWh; at an explicitly assumed 80 W for 480 s, 0.01067 kWh. Add valves/electronics. Actual tariffs, installation and maintenance still need quotes.

Reference calibration, sensor saturation, humidity and mist interference, clean inlets/filters and independent site events are pilot gates. EPA sensor guidance [10]. No physical design or firmware is validated by this packet.

05 / TEN-MINUTE STORY

Lead with the demo. Keep the evidence honest.

Time	Show / say
0:00-0:45	Introduce CTRL_V and the software prototype
0:45-1:45	Local problem; continuous/reactive approaches; proposed contribution
1:45-2:45	Dataset, causal task, split and laboratory limitation
2:45-4:15	Live measured replay, inputs, baselines and later actual
4:15-5:15	Mixed held-out errors and warning failures
5:15-6:45	Diagonal site replay, wind shift and sensor loss
6:45-7:45	Water/exposure tradeoff and shared assumptions
7:45-9:00	Proposed architecture, costs and pilot validation
9:00-10:00	Completed software, team and request for pilot feedback

Questions the team should prepare for

Why use the model if the mean wins? The mean is a credible baseline. This model is a preliminary learned demonstration; lower RMSE does not establish universal superiority. New model research needs new untouched evaluation data.

Does it predict a real boundary? No. It forecasts the measured lab monitor; spatial mapping and misting response are separate, labelled assumptions.

How much water will a site save? Unknown. Compare actual flow and exposure under comparable independent activities before claiming field or daily savings.

Where is the hardware? Deferred. The team must explicitly authorize a calibrated Round 2 pilot. Qualification and feedback are not inferred.

What if the service fails? Use the labelled saved inference replay and PDF/screenshots. Do not describe saved output as fresh model execution.

The full speaking script is included as docs/judge-script.md and downloadable from Present. Practice spoken delivery with a real timer and check projector readability. The automated rehearsal verifies software only.

06 / SOURCE AND REPRODUCIBILITY

Sources you can inspect.

[1] World Bank / local construction and traffic context

<https://blogs.worldbank.org/en/endpovertyinsouthasia/bangladesh-urgent-call-clean-air>

[2] Askarov and Choi 2024 / Mendeley V1 / CC BY 4.0

<https://data.mendeley.com/datasets/7f22n9v7hp/1>

[3] DFRobot / SEN0177 PM monitor

<https://www.dfrobot.com/product-1272.html>

[4] DFRobot / ESP32-E DFR0654

<https://www.dfrobot.com/product-2195.html>

[5] DFRobot / SEN0483 wind speed

<https://www.dfrobot.com/product-2339.html>

[6] DFRobot / SEN0482 wind direction

<https://www.dfrobot.com/product-2340.html>

[7] TechShopBD / pump set

<https://techshopbd.com/product/high-pressure-single-motor-water-pump-set-fl-4200>

[8] TechShopBD / 12 V valve

<https://techshopbd.com/product/solenoid-valve-12v-34-inch/>

[9] TechShopBD / four-channel relay

<https://techshopbd.com/product/relay-module-4-channel-12v>

[10] US EPA / sensor limitations and calibration

<https://www.epa.gov/air-sensor-toolbox/frequent-questions-about-air-sensors>

Dataset reuse

Derived replay, causal preparation, forecasts and audit/evaluation plots credit Komiljon Askarov and Jae-ho Choi (2024), DOI 10.17632/7f22n9v7hp.1, CC BY 4.0. Originals are unchanged. Gridding/features/forecasts are project transformations. No contributor endorsement. The rejected 2020 data's separate attribution appears in README.md.

Local evidence

models/model-card.md; reports/evaluation/test-metrics.json; compressed full test traces; reports/training/validation-selection.json; reports/simulation/summary.json; configs/forecast-task.json; configs/feasibility.json; data/manifest.json; reports/readiness/.

Public source: github.com/arifshekhk8/DustTwin-AI

The website was implemented originally; no code was imported from the reference frontend, whose reuse license was not recorded. All runtime evidence needed for the demonstration is included locally.